Extra Dessert #5: Safety, RF Exposure, and Grounding

FCC RF exposure rules, MPE evaluation, SAR, microwave hazards, tower climbing safety, and station grounding.

Group E0A · 12 pool questions · Self-study reference · Practice exams at eham.net/exams

RF Exposure And MPE

Question	Answer	Q
Evaluating exposure at a neighbor's home	Ensure signals are less than the **uncontrolled MPE limits** (since the neighbor cannot control or know about your station)	E0A02
Most restrictive MPE frequency range	**30-300 MHz** (body resonant region)	E0A03
Why separate E and H limits below 300 MHz	Body responds to both fields; ground reflections vary field strength with location; E and H peaks can occur at different places – **all of the above**	E0A06
Multi-transmitter sites: who must mitigate?	Each transmitter that produces **5% or more of its MPE limit** in areas where the total MPE limit is exceeded	E0A04
When evaluation must be done on 80 m	**An evaluation must always be performed.** (All amateur stations need an exposure evaluation under current FCC rules.)	E0A10
What equipment is exempt?	**Hand-held transceivers sold before May 3, 2021**	E0A09
SAR	Specific Absorption Rate: **the rate at which RF energy is absorbed by the body**	E0A08
Microwave hazard	**High-gain antennas can result in high exposure levels** (concentrated beams)	E0A05

Tower Safety

Question	Answer	Q
"100% tie-off" meaning	**At least one lanyard attached to the tower at all times** (you never disconnect both at once)	E0A07
What to attach lanyards to while climbing	**Tower legs** (the strongest structural members)	E0A11
Where to attach shock-absorbing lanyard above ground	**Above the climber's head level** (shortens the fall distance before the lanyard stops you)	E0A12

Grounding

Question	Answer	Q
Primary function of an external ground rod	**Lightning charge dissipation**	E0A01

Dessert 5 Question Bank (E0A)

E0A01. What is the primary function of an external earth connection or ground rod?
A. Prevent static build up on power lines.
B. Lightning charge dissipation.
C. Reduce RF current flow between pieces of equipment.
D. Protect breaker panel from power surges.

E0A02. When evaluating RF exposure levels from your station at a neighbor's home, what must you do?
A. Ensure signals from your station are less than the controlled maximum permissible exposure (MPE) limits.
B. Ensure signals from your station are less than the uncontrolled maximum permissible exposure (MPE) limits.
C. Ensure signals from your station are less than the controlled maximum permissible emission (MPE) limits.
D. Ensure signals from your station are less than the uncontrolled maximum permissible emission (MPE) limits.

E0A03. Over what range of frequencies are the FCC human body RF exposure limits most restrictive?
A. 300 kHz - 3 MHz.
B. 3 - 30 MHz.
C. 30 - 300 MHz.
D. 300 - 3000 MHz.

E0A04. When evaluating a site with multiple transmitters operating at the same time, the operators and licensees of which transmitters are responsible for mitigating over-exposure situations?
A. Each transmitter that produces 20 percent or more of its MPE limit in areas where the total MPE limit is exceeded.
B. Each transmitter operating with a duty cycle greater than 25 percent.
C. Each transmitter that produces 5 percent or more of its MPE limit in areas where the total MPE limit is exceeded.
D. Each transmitter operating with a duty cycle greater than 50 percent.

E0A05. What hazard is created by operating at microwave frequencies?
A. Microwaves are ionizing radiation.
B. The high gain antennas commonly used can result in high exposure levels.
C. Microwaves are in the frequency range where wave velocity is higher.
D. The extremely high frequency energy can damage the joints of antenna structures.

E0A06. Why are there separate electric (E) and magnetic (H) MPE limits at frequencies below 300 MHz?
A. The body reacts to electromagnetic radiation from both the E and H fields.
B. Ground reflections and scattering cause the field strength to vary with location.
C. E field and H field radiation intensity peaks can occur at different locations.
D. All these choices are correct.

E0A07. What is meant by "100% tie-off" regarding tower safety?
A. All loose ropes and guys secured to a fixed structure.
B. At least one lanyard attached to the tower at all times.
C. All tools secured to the climber's harness.
D. All circuit breakers feeding power to the tower must be tied closed with tape, cable, or ties.

E0A08. What does SAR measure?
A. Signal attenuation ratio.
B. Signal amplification rating.
C. The rate at which RF energy is absorbed by the body.
D. The rate of RF energy reflected from stationary terrain.

E0A09. Which of the following types of equipment are exempt from RF exposure evaluations?
A. Transceivers with less than 7 watts of RF output.
B. Antennas that radiate only in the near field.
C. Hand-held transceivers sold before May 3, 2021.
D. Dish antennas less than one meter in diameter.

E0A10. When must an RF exposure evaluation be performed on an amateur station operating on 80 meters?
A. An evaluation must always be performed.
B. When the ERP of the station is less than 10 watts.
C. When the station's operating mode is CW.
D. When the output power from the transmitter is less than 100 watts.

E0A11. To what should lanyards be attached while climbing?
A. Antenna mast.
B. Guy brackets.
C. Tower rungs.
D. Tower legs.

E0A12. Where should a shock-absorbing lanyard be attached to a tower when working above ground?
A. Above the climber's head level.
B. To the belt of the fall-arrest harness.
C. Even with the climber's waist.
D. To the next lowest set of guys.

Answer Key

E0A01: B	E0A02: B	E0A03: C	E0A04: C
E0A05: B	E0A06: D	E0A07: B	E0A08: C
E0A09: C	E0A10: A	E0A11: D	E0A12: A